Shad Abdullah

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Education

B.Sc. in Computer Science and Engineering, North South University Jan 2019 – Feb 2024
Dhaka, Bangladesh

Experience

Backend Engineer, Brainicon Technology Apr 2025 – Present

- Design and build scalable REST APIs for multi-tenant SaaS products using Django, Django REST Framework and PostgreSQL.
- Own backend architecture end-to-end: data modeling, JWT authentication, role-based access control, and Dockerized development and deployment.
- Collaborate using Git-based workflows and write API documentation for frontend and mobile teams.

Data Analytics Intern, eSystems Research and Development Lab Sep 2023 – Dec 2023

- Completed a 4-month industrial training program (ICT Division, Government of Bangladesh research grant) on data analytics.
- Built data pipelines and a web application with Python, SQL, Pandas and Flask; visualized insights with Matplotlib for data-driven decisions.
- Scored 100% on task completion, communication and coding assessments.

Projects

Agami EMS – Multi-Tenant Education Management SaaS 2025

- Architected a multi-tenant SaaS platform from scratch using django-tenants with isolated public and per-tenant PostgreSQL schemas, serving multiple institutions from a single codebase.
- Implemented a custom group-aware JWT authentication layer, tenant/portal context switching, Google OAuth, reCAPTCHA, and an email-based invitation and role-assignment system.
- Built core modules: Campus, People, Exams (marks calculation, grouping, publish/lock academic periods) and Finance (income, expense and debt ledger).
- Containerized the stack with Docker and Docker Compose; served via Gunicorn and WhiteNoise.

Medical Insurance Cost Prediction – Machine Learning Project 2024

- Developed a machine learning model using Random Forest Regression to predict medical insurance costs with 90% accuracy.
- Improved model performance through hyperparameter tuning using Python, Pandas and Scikit-learn.

Publication

Identifying Threats on Social Media to Spot Offensive Behavior (IEEE) 2024

- Fine-tuned DistilBERT, m-BERT, XLM-RoBERTa and BanglaBERT on 44,000 Bangla social-media comments; achieved 0.83 accuracy (XLM-RoBERTa) and 0.85 AUC (m-BERT).
- Paper: https://ieeexplore.ieee.org/document/10705230

Skills

Languages: Python, SQL, C
Backend: Django, Django REST Framework, django-tenants, JWT, Flask, REST APIs
Databases: PostgreSQL, MySQL, pgAdmin
DevOps and Tools: Docker, Docker Compose, Git, Gunicorn, WhiteNoise, VS Code, PyCharm
ML and Data: Pandas, Matplotlib, Seaborn, Scikit-learn, Machine Learning, NLP